Colby Okazaki

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Education

University of California, Santa Cruz — Santa Cruz, California
B.S. in Computer Science — *Expected Graduation, Jun 2028*

- **GPA:** 3.66/4.00, *Dean's Honors*
- **Courses:** Data Structures and Algorithms, Machine Learning: Data Analysis, Computer Systems and C Programming, Programming Abstractions: Python, Computer Systems and Assembly Language, Applied Discrete Mathematics, Linear Algebra, Vector Calculus, Principles of Computer Systems Design, Mathematical Thinking for Computer Science

Experience

Blueprint — Santa Cruz, California
Software Developer — *Oct 2026 – Present*

- Joined a team building a cross-platform mobile application with **React Native** and **TypeScript** for a local nonprofit, using **Expo** and **Supabase**
- Collaborating with a cross-functional team of developers and designers, owning and shipping individual features for a local nonprofit client

Technical Skills

Languages: Python, Java, C++, C, Swift, JavaScript, TypeScript, SQL (PostgreSQL), HTML/CSS
Frameworks & Tools: React, Next.js, Node.js, Express, Socket.io, Konva.js, Jest, React Testing Library, Prisma, OpenCV, PyTorch, REST APIs, Git, VS Code, Cursor, Claude Code, Vercel
Operating Systems: macOS, Linux

Projects

Collaborative Whiteboard | *React, Konva.js, Socket.io, Node.js, Express, Jest* — Oct 2026 – Present

- Built a real-time collaborative whiteboard in **React** and **Konva.js** where multiple users draw on a shared canvas, with a **Socket.io** backend syncing strokes across clients in room-based boards
- Implemented live multi-user cursors with a custom throttling utility that caps updates at 50ms while guaranteeing the final position is delivered, cutting network traffic during mouse movement
- Sent a full board snapshot to users joining mid-session so every client starts in the same state, with **Jest** and **React Testing Library** tests covering server state, timing logic, and UI components

Sports Stats Query Engine | *TypeScript, Next.js, React, REST APIs, NDJSON Streaming* — Oct 2026

- Built a sports stats search engine in **Next.js** and **TypeScript** that answers plain-English questions with a one-sentence answer and stats table across 7 leagues, using ESPN's public **APIs**
- Designed a rule-based natural-language parser that extracts league, player/team, stat, season and question type, backed by 56 stat definitions with synonyms, streaming results as **NDJSON**
- Shipped live scoreboards with in-game leaders, polling **Gamecast** pages with win probability and play-by-play, and full box scores, using server-side caching to limit ESPN requests

Robotic Ring Sorter | *Python, OpenCV, PyTorch, faster-whisper, SO-101 Robot Arm* — Oct 2026

- Built a voice-controlled robot arm at the **AMD × LeRobot hackathon** that identifies colored rings via an overhead camera and picks and places them using an SO-101 arm, with local speech-to-text through **faster-whisper**
- Implemented hue-based color detection in **OpenCV** that stayed robust in direct sunlight, and a grab-pose system that blends recorded human demonstrations with a taught position grid

Movie Ticket Booking | *TypeScript, Node.js, Express, Prisma, PostgreSQL, React, JWT* — Jul 2026

- Built a full-stack cinema booking app with **JWT** auth, role-based user/admin portals, and a **React** client for browsing showtimes, seat selection, and ticket management
- Modeled movies, showtimes, and seat inventory in **Prisma/PostgreSQL**, implementing transactional seat booking with conflict handling to prevent double-bookings
- Built an admin operations dashboard with occupancy/revenue reporting for movies, showtimes, and reservations, seeded with a demo catalog of real film posters and multiplex-style showtimes